

KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

(Autonomous)

Avinashi Road, Arasur, Coimbatore 641 407

*Learn Beyond***Course Code & Title:** U21ECG04 - Internet of Things and Its Applications**CIAT:** 1**Year:** II **Dept:** AD **Ac Yr:** 2026-2027**Semester:** 05**Duration:** 90 Minutes**Maximum Marks:** 60**Section-A (10X1=10 Marks)****Answer All Questions**

Q. No	Question	RBT	CO
1	An embedded system is primarily designed to perform _____. a) General-purpose computing b) A dedicated application c) Scientific computation d) Office automation	R	CO1
2	The processor in an embedded system mainly performs _____. a) Power regulation b) Data processing and control c) Signal transmission d) Data storage	R	CO1
3	A simple embedded system generally does NOT require _____. a) Microcontroller b) Sensor c) Real-Time Operating System d) Power Supply	R	CO1
4	A smartphone simultaneously handles voice calls, GPS navigation and multimedia playback. Which hardware block coordinates these operations? a) Power Management Unit b) CPU/System-on-Chip (SoC) c) Battery d) Touchscreen	U	CO1
5	_____ microcontroller is used in the Arduino Uno board. a) ATmega2560 b) ATmega32U4 c) ATmega328P d) ESP32	U	CO2
6	_____ function executes continuously during the operation of an Arduino program. a) setup() b) execute() c) loop() d) begin()	U	CO2
7	What is the main purpose of PWM pins in Arduino? a) To read digital signals b) To simulate varying voltages between 0 and 5V c) To store program code d) To measure resistance	R	CO2
8	An engineer is developing a smart classroom attendance system that communicates with a computer as a Human Interface Device (HID). Which type of interface is most suitable for this application? a) Arduino Uno b) Arduino Nano c) Arduino Mega 2560 d) Arduino Leonardo	U	CO2
9	_____ is commonly used to measure the surrounding temperature in Arduino-based embedded systems. a) MQ2 Sensor b) DHT11 Sensor c) Ultrasonic Sensor d) PIR Sensor	R	CO3
10	_____ sensor is commonly used to measure the distance between an object and the sensor. a) DHT11 b) MQ2 c) HC-SR04 Ultrasonic d) PIR Sensor	R	CO3

Section-B (10X2=20 Marks)**Answer All Questions**

Q. No	Question	RBT	CO
11	List four important characteristics of an embedded system.	R	CO1
12	Mention the role of an Interrupt Service Routine (ISR) in the embedded system.	U	CO1
13	An automatic street lighting system switches ON the lights after sunset. Mention the sequence of operation using the embedded system model.	U	CO1
14	Draw the flow of information from the soil moisture sensor to the water pump actuator in a smart irrigation system.	U	CO1
15	Justify the different Arduino family boards and mention one application of each.	U	CO2
16	Mention any four digital and analog I/O functions available in Arduino.	R	CO2
17	Justify the use of the Arduino Mega 2560 in a smart parking system with multiple sensors.	R	CO2
18	Identify the Arduino function used to perform repeated pick-and-place operations in a robotic arm.	R	CO2
19	Justify the use of a temperature sensor in an Arduino-based greenhouse monitoring system.	R	CO3
20	Identify the purpose of an ultrasonic sensor in a water level monitoring system.	R	CO3

Section-C (2X12=24 Marks & 1X6=6 Marks)**Answer All Questions**

Q. No	Question	RBT	Marks	CO
21 a)	Smart home automation systems integrate lighting, security and environmental monitoring into a single embedded platform. Illustrate the architecture of a Microcontroller Unit (MCU) with a neat block diagram and describe the functions of its major internal components.	U	12	CO1
(Or)				
21 b)	Smart greenhouse systems continuously monitor environmental conditions and automatically regulate cooling fans and sprinklers to improve crop productivity. Outline the embedded system model used in this application and identify the roles of sensors, processor, memory and actuators.	U	12	CO1
22 a)	Traffic signal controllers repeatedly execute predefined sequences while responding to changing traffic conditions. Illustrate the functions of setup(), loop(), if-else statements, delay() and digital I/O functions used in developing this application.	U	12	CO2
(Or)				
22 b)	USB-based attendance systems identify users by emulating a keyboard when connected to a computer. Interpret the architecture, features and applications of Arduino Leonardo, highlighting its suitability for this application.	U	12	CO2
23 a)	A residential building requires an automatic system to monitor the water level in an overhead tank and control the pump. Develop an Arduino-based solution using a water-level sensor, relay module, and water pump, and prepare a suitable interfacing diagram.	Ap	6	CO3
(Or)				
23 b)	A smart home requires an automated security system to detect unauthorized human movement. Develop an Arduino-based interfacing circuit using a PIR sensor and suitable output devices for intrusion detection.	Ap	6	CO3